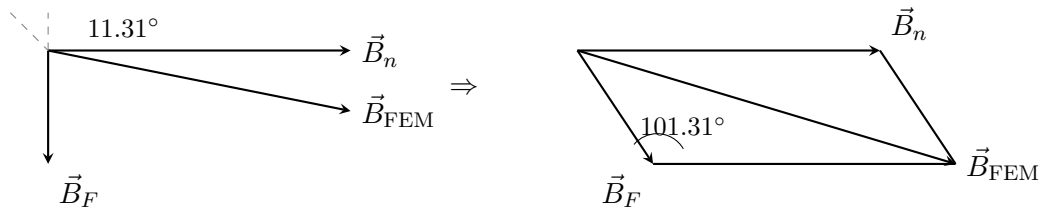


Vector diagram:



By the Law of Cosines:

$$\cos(101.31^\circ) = \frac{\vec{B}_n^2 + \vec{B}_F^2 - \vec{B}_{\text{FEM}}^2}{2 \cdot \vec{B}_n \cdot \vec{B}_F} = \frac{135.98^2 + \vec{B}_F^2 - 136.4^2}{2 \cdot 135.98 \cdot \vec{B}_F}$$

$$\Rightarrow \boxed{\vec{B}_F = 2.065 \text{ mT}}$$